No.

Date.

Homework #1

习题= 3, 4, 9, 10, 14, 16

T3 (1) $(A,B)=(0,0)\Rightarrow X(t)=0,\ t\geq 0$; $(A,B)=(0,1)\Rightarrow X(t)=1,\ t\geq 0$

$(A,B)=(0,-1)\Rightarrow X(t)=-1,\ t\geq 0$;

$(A,B)=(1,0),(1,1),(1,-1)\Rightarrow X(t)=t,\ t\geq 0$

$(A,B)=(-1,0);(-1,1);(-1,-1)\Rightarrow X(t)=-t,\ t\geq 0$

(2) $P(X(1)=1)=\frac{4}{9}$, $P(X(2)=1)=\frac{1}{9}$

$P(X(1)=1, X(2)=1)=\frac{1}{9}$.

T4 (1) $A=0\Rightarrow Z(t)=1,\ t\geq 0$

$A=1\Rightarrow Z(t)=Xt,\ t\geq 0$

$P(Z(1)<1)=\frac{1}{2}P(X<1)=\frac{1}{4}$

$P(Z(2)<2)=\frac{1}{2}+\frac{1}{2}P(X<1)=\frac{3}{4}$

$P(Z(1)<1, Z(2)<2)=\frac{1}{2}P(X<1)=\frac{1}{4}$

(2) $\mu_Z(t)=E(Z(t))=\frac{1}{2}t+\frac{1}{2},\ t\geq 0$

$R_Z(s,t)=E(Z(s)Z(t))=E(A^2Xst+AX(1-A)(s+t)+(1-A)^2)$

$E(AB)=E(A)=\frac{1}{2}$, $E(X^2)=Var(X)+[E(X)]^2=2$

故 $R_Z(s,t)=st+\frac{1}{2}$

CS 扫描全能王
3亿人都在用的扫描App

Date

T9 $X(t)=At+B$, $t>0$, $E(A)=\mu$, $D(A)=\sigma^2>0$

$E(A^2)=[E(A)]^2+D(A)=\mu^2+\sigma^2$

(1) $\mu_X(t)=E(X(t))=\mu t+1$

$R_X(s,t)=E(X(s)X(t))=E(A^2ts+AB(s+t)+B^2)$

$=(\mu^2+\sigma^2)ts+\mu^2(s+t)+\mu^2+\sigma^2$

$C_X(s,t)=R_X(s,t)-\mu_X(t)\mu_X(s)=\sigma^2ts+\sigma^2$

(2) $A\sim N(0,1)$ $At\sim N(0,t^2)$

$At+B\sim N(0,t^2+1)$ $\therefore \{X(t)\}$是正态过程

$X(t)\sim N(0,t^2+1)$

$X(t)-X(s)=A(t-s)$ $X(t)-X(s)\sim N(0,(t-s)^2)$

$X(t)+X(s)=A(t+s)+2B$ $X(t)+X(s)\sim N(0,(t+s)^2+4)$

T10 $X_0,X_1,\cdots$独立同分布 $P(X_0=1)=p=1-P(X_0=0)$ $0<p<1$

令$Y_n=X_n+X_{n+1}+X_{n+2}$

(1)

Y_0	0	1	2	3
p	$(1-p)^3$	$3(1-p)^2p$	$3p^2(1-p)$	p^3

(2) $P\{Y_1=1\mid Y_0=2\}=\frac{2}{3}(1-p)$ $P\{Y_1=2\mid Y_0=2\}=\frac{2}{3}p+\frac{1}{3}(1-p)=\frac{1}{3}(1+p)$

$P\{Y_1=3\mid Y_0=2\}=\frac{1}{3}p$

Y_1	1	2	3
P	$\frac{2}{3}(1-p)$	$\frac{1}{3}(1+p)$	$\frac{1}{3}p$

CS 扫描全能王
3亿人都在用的扫描App

No.

Date.

(3) $P(Y_0=1, Y_1=0, Y_4=1)$

$= P(X_0=1, X_1=X_2=X_3=0, X_4=1)$

$= p^2(1-p)^3$

(4) $\mu_Y(n) = E(Y(n)) = 3p$

$C_Y(t,s) = R_Y(t,s) - \mu_Y(t)\mu_Y(s)$

$R_Y(t,s) = E(Y_t Y_s)$

$= E[(X_t+X_{t+1}+X_{t+2})(X_s+X_{s+1}+X_{s+2})]$

$$= \begin{cases} 9p^2 & s>t-2 \text{ 或 } s<t+2 \\ 8p^2+p & s=t-2 \text{ 或 } s=t+2 \\ 7p^2+2p & s=t+1 \text{ 或 } s=t+1 \\ 6p^2+3p & s=t \end{cases}$$

故 $C_Y(t,s) = \begin{cases} 0 & s>t-2 \text{ 或 } s>t+2 \\ p-p^2 & s=t-2 \text{ 或 } s=t+2 \\ 2p-2p^2 & s=t-1 \text{ 或 } s=t+1 \\ 3p-3p^2 & s=t \end{cases}$

CS 扫描全能王
3亿人都在用的扫描App

No. 6

Date.

T14 $\{X(t), t\in T\}$ 与 $\{Y(t), t\in T\}$ 不相关

$Z(t)=a(t)X(t)+b(t)Y(t)+c(t)\quad t\in T$

$\mu_Z(t)=E(Z(t))=a(t)\mu_X(t)+b(t)\mu_Y(t)+c(t)$

$C_Z(s,t)=R_Z(s,t)-\mu_Z(s)\mu_Z(t)$

$=a(t)a(s)C_X(s,t)+b(t)b(s)C_Y(s,t)$

T16 $\{X(t), t\in(-\infty,+\infty)\}$ 与 $\{Y(t), t\in(-\infty,+\infty)\}$ 相互独立

已知 $\mu_X(t)$, $R_X(t,s)$, $\mu_Y(t)$, $R_Y(t,s)$

$Z(t)=X(t)Y(t)$

$\mu_Z(t)=\mu_X(t)\mu_Y(t)$

$R_Z(s,t)=E[X(s)X(t)Y(s)Y(t)]=R_X(t,s)R_Y(t,s)$

$R_{XZ}(s,t)=E[X(s)X(t)Y(t)]=R_X(t,s)\mu_Y(t)$

CS 扫描全能王
3亿人都在用的扫描App